



Phase III: Software Design and Modeling

CEN 302 – Software Engineering

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1. System Architecture

The Thesis Progress Tracker follows a three-tier architecture consisting of a user interface, a backend and a database. The frontend is built using HTML/CSS and runs on the web server, generating the pages that users see in their browser. It sends requests to the backend, which is also developed using Python. The backend processes all requests, enforces role-based access control, handles file uploads, and communicates with the database. The database built with SQL stores all persistent data including user accounts, thesis submissions, feedback, messages, deadlines, and notifications. To explain in more detail, we can take 3 different examples:

1. Student Submits Thesis Progress

- Student uploads thesis file (PDF/DOCX) and clicks Submit (Frontend – HTML form)
- Request is sent to the Backend
- Backend performs validations and conditions:
 - Checks if file is PDF/DOCX
 - Checks file size ($\leq 10\text{MB}$)
 - Verifies student is logged in
- Backend sends request to Database:
 - Stores file path
 - Saves submission details: student_id, description, timestamp, status = *pending*
- Database confirms storage
- Backend sends success response to Frontend
- Student sees confirmation message and updated submission list
- System triggers notification for Supervisor as new submission.

2. Supervisor Reviews and Gives Feedback

- Supervisor opens Student Submissions dashboard (Frontend)
- Request goes to Backend
- Backend queries Database:
- Retrieves submissions only for assigned students and fetches: file, submission date and current status
- Backend sends data to Frontend
- Supervisor selects a submission and adds feedback

- Request goes to Backend
- Backend saves feedback in Feedback table:
- Updates submission status: approved or rejected
- Backend sends update to Database
- Database stores: feedback comment, supervisor_id and submission_id
- Backend sends response to Frontend
- Student immediately sees feedback and updated status
- System sends notification to Student

3. Admin Assigns Supervisor to Student

- Admin selects student and assigns supervisor (Frontend dashboard)
- Request goes to Backend
- Backend verifies admin role (authorization check)
- Backend updates Database: Links student_id with supervisor_id
- Database confirms update
- Backend sends response to Frontend
- Assignment is visible in both Student and Supervisor dashboards

1. Component Diagram

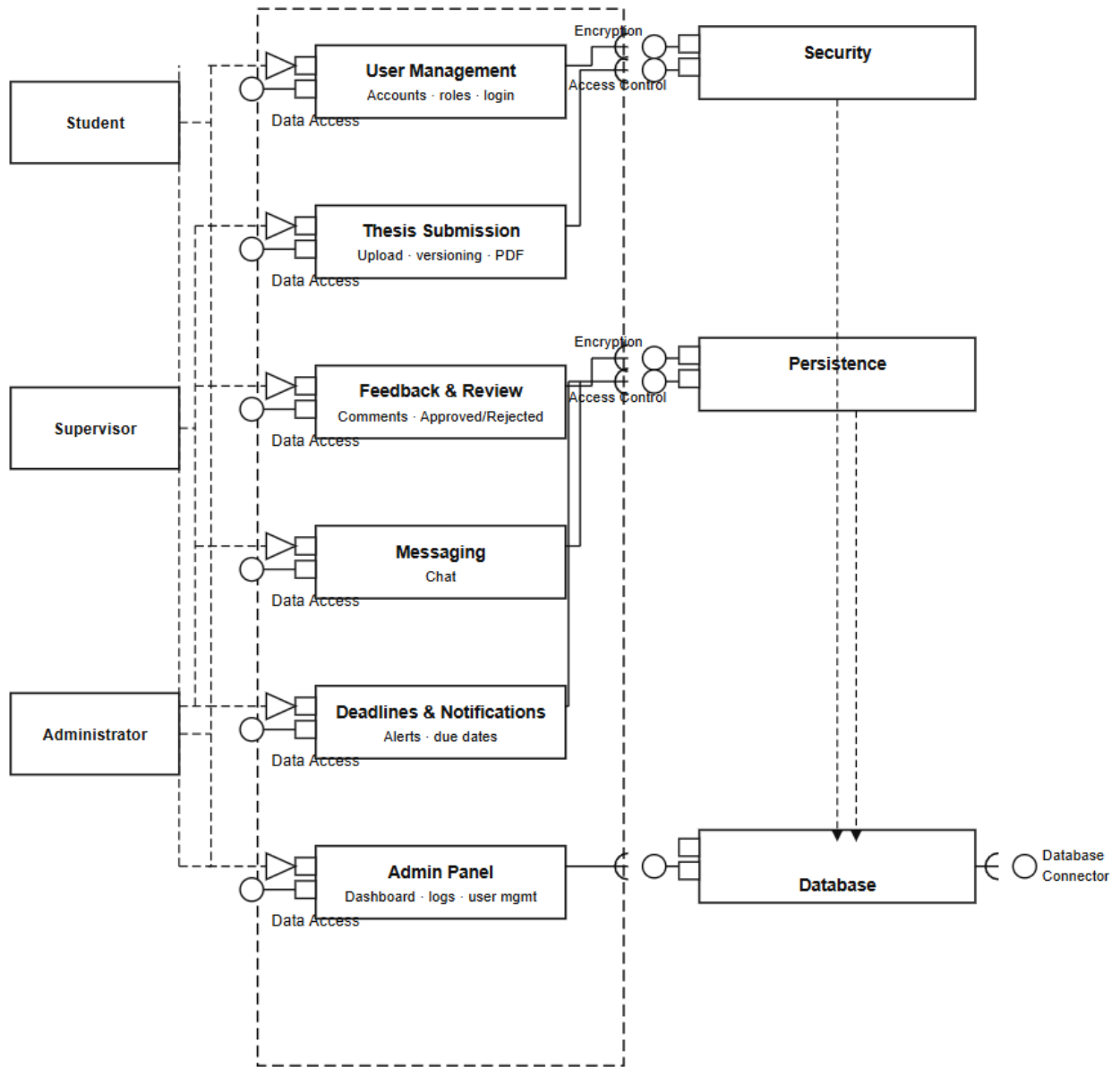


Figure 1: UML Component Diagram

Actors (left side):

Student: the primary user who submits thesis documents, views deadlines, reads feedback, and communicates with their supervisor

Supervisor: guides and evaluates students by reviewing submissions, providing feedback, assigning deadlines, and exchanging messages

Administrator: manages the system by handling user accounts, assigning supervisors to students, and monitoring overall system activity

Components (middle part):

User Management: handles account creation, login authentication, and role assignment for all three user types

Thesis Submission: manages PDF uploads, stores each submission as a new version, and tracks review status per submission

Feedback & Review: allows supervisors to write comments and set submission status to Approved or Rejected, immediately visible to the student

Messaging: enables text communication between students and supervisors, supporting one PDF attachment per message

Deadlines & Notifications: manages deadline creation and assignment by supervisors, and delivers in-app alerts to relevant users when events occur

Admin Panel: provides administrators with user management tools, supervisor-student assignments, a system dashboard, and access to the full activity log

Data Access: meaning it reads/writes all persistent data, accounts, submissions, feedback, messages, deadlines, and notifications from the database using parameterised queries to prevent SQL injection

External components (right side):

Security: enforces login authentication, password encryption, and role-based access control to ensure each user can only access permitted data

Persistence: handles reliable file storage for thesis documents, maintains submission version history, and ensures data consistency and automatic backups

Database: the MySQL relational database that stores all system data across nine tables including Students, Supervisors, Submissions, Feedback, Messages, Deadlines, and Notifications, exposed through a Database Connector interface

2. Detailed Design

1. UML Class Diagram

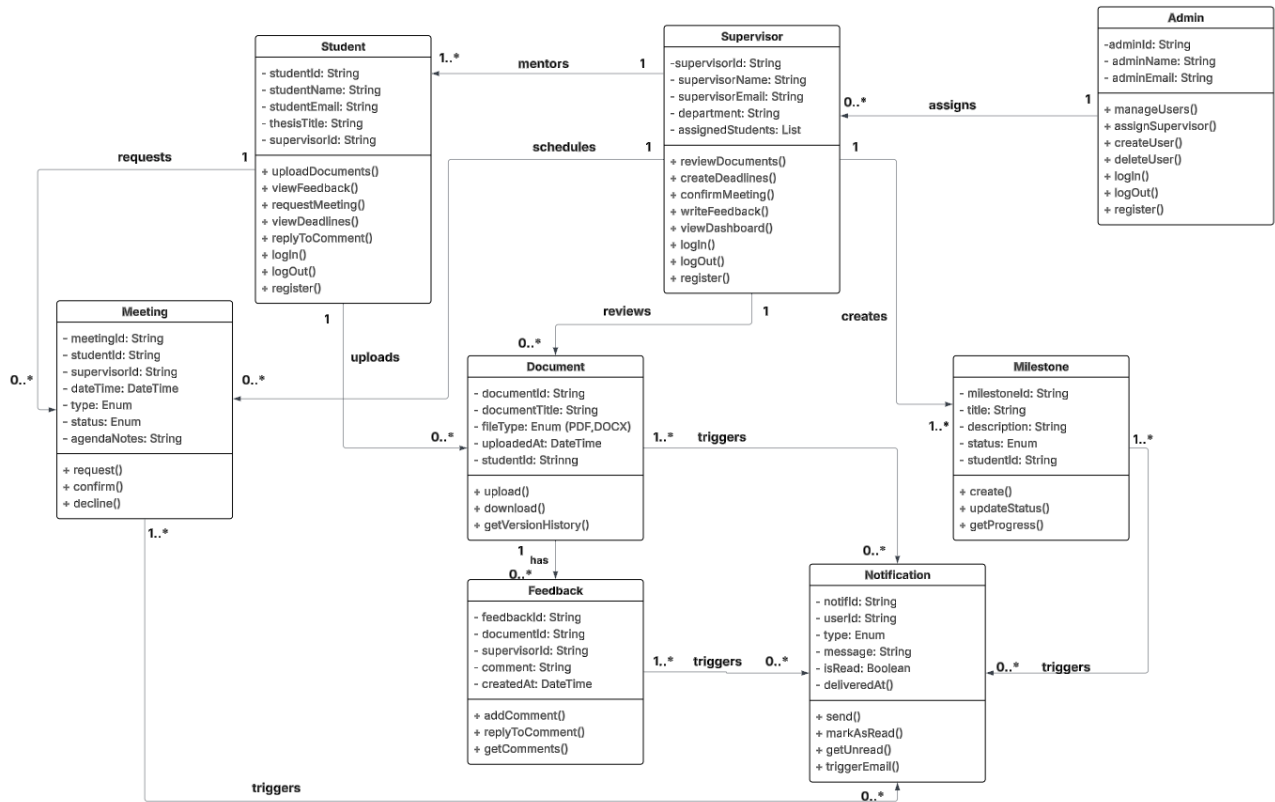


Figure 2: UML Class Diagram

(Note*: A clearer version of the diagram can be accessed via the following link)

Diagram Link: https://lucid.app/lucidchart/d3d57928-fac0-46d3-9091-56ce7eabbaf3/edit?viewport_loc=-1365%2C-792%2C4955%2C2275%2C0_0&invitationId=inv_6e55443e-59a7-4aec-a4f4-fc1a2f9bcc7c

2. UML Sequence Diagrams

a. Student Uploads Thesis

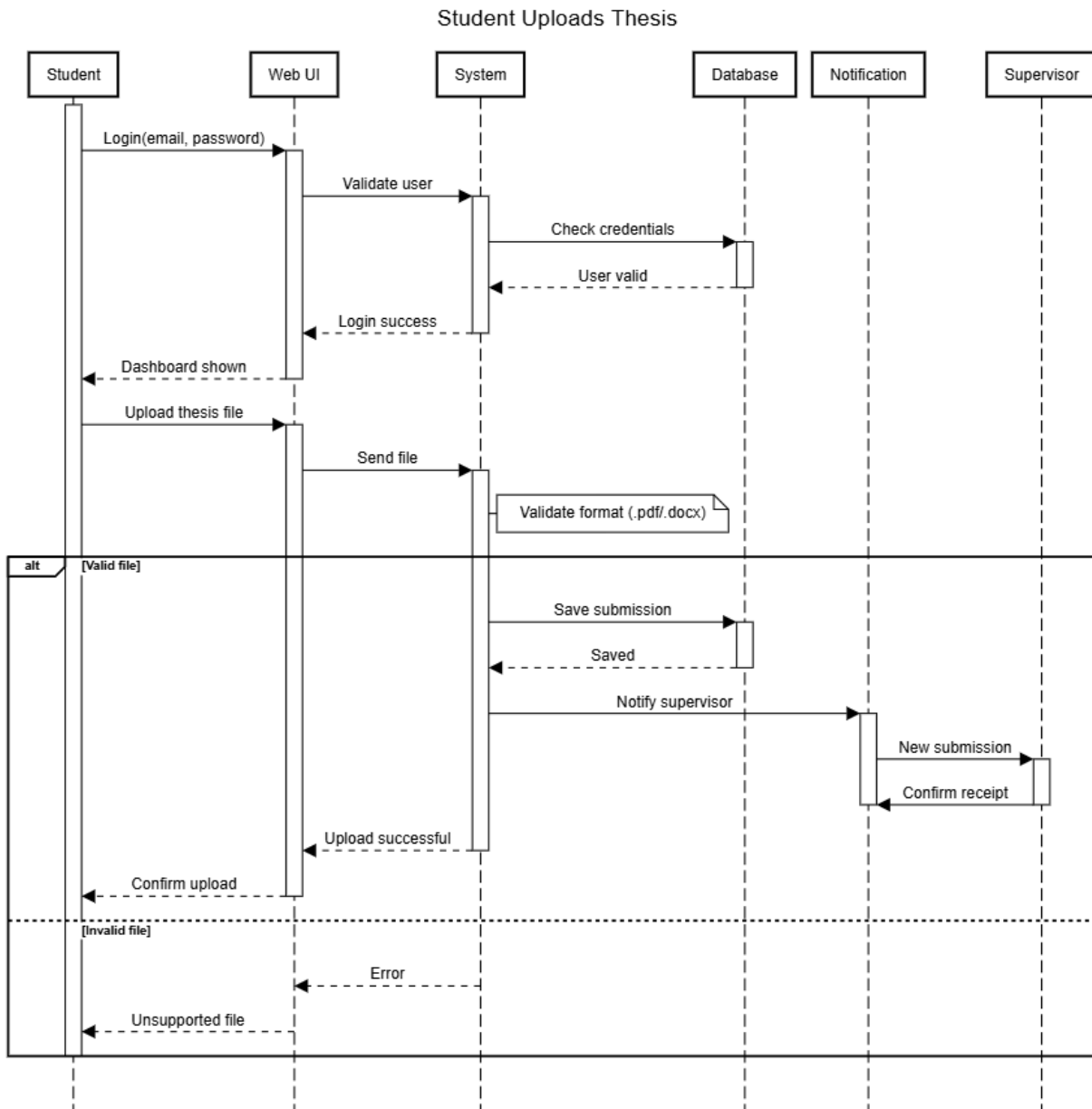


Figure 3: Sequence Diagram - Student Uploads Thesis

b. Supervisor Reviews Thesis

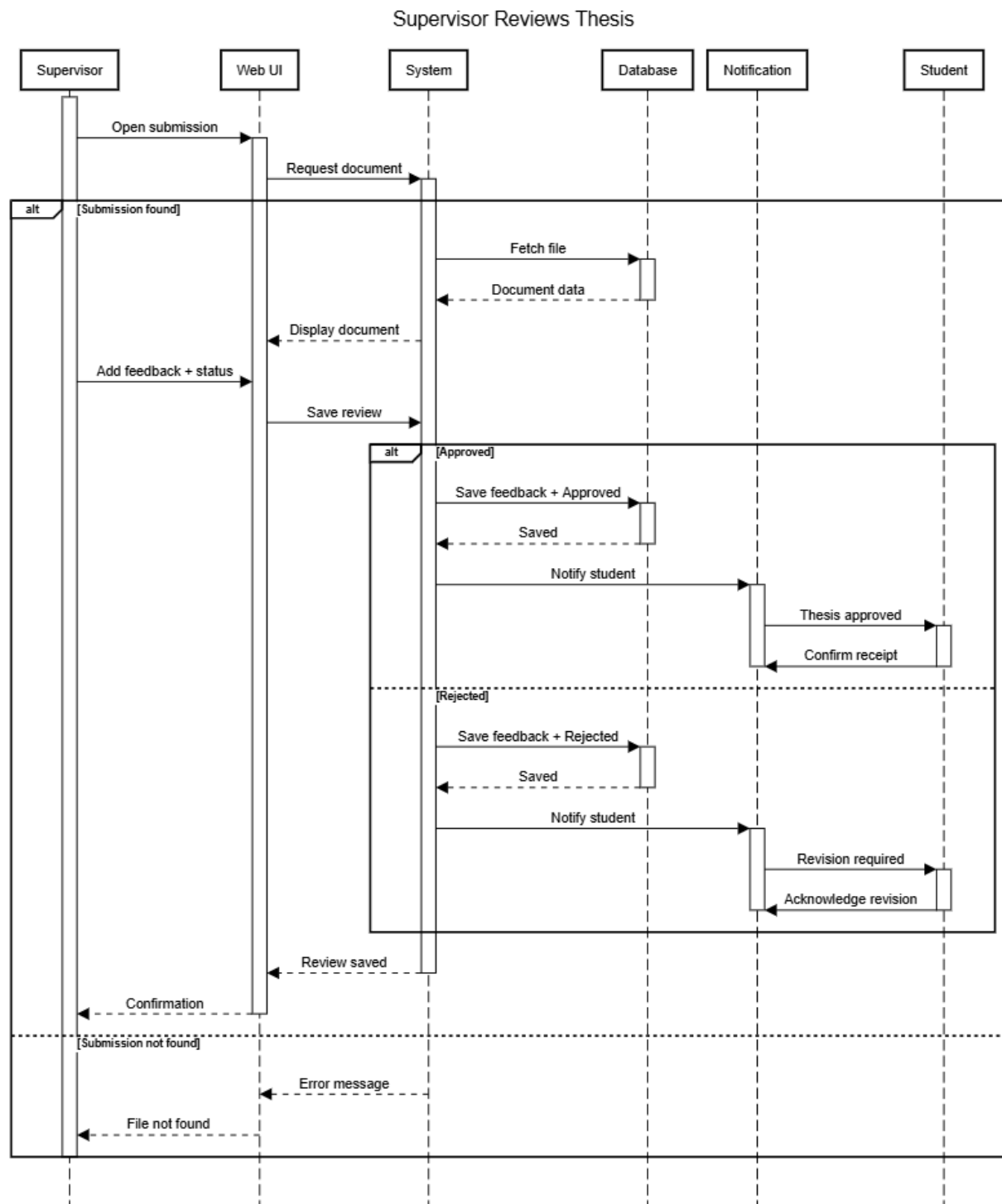


Figure 4: Sequence Diagram - Supervisor Reviews Thesis

c. Supervisor Schedules Meeting

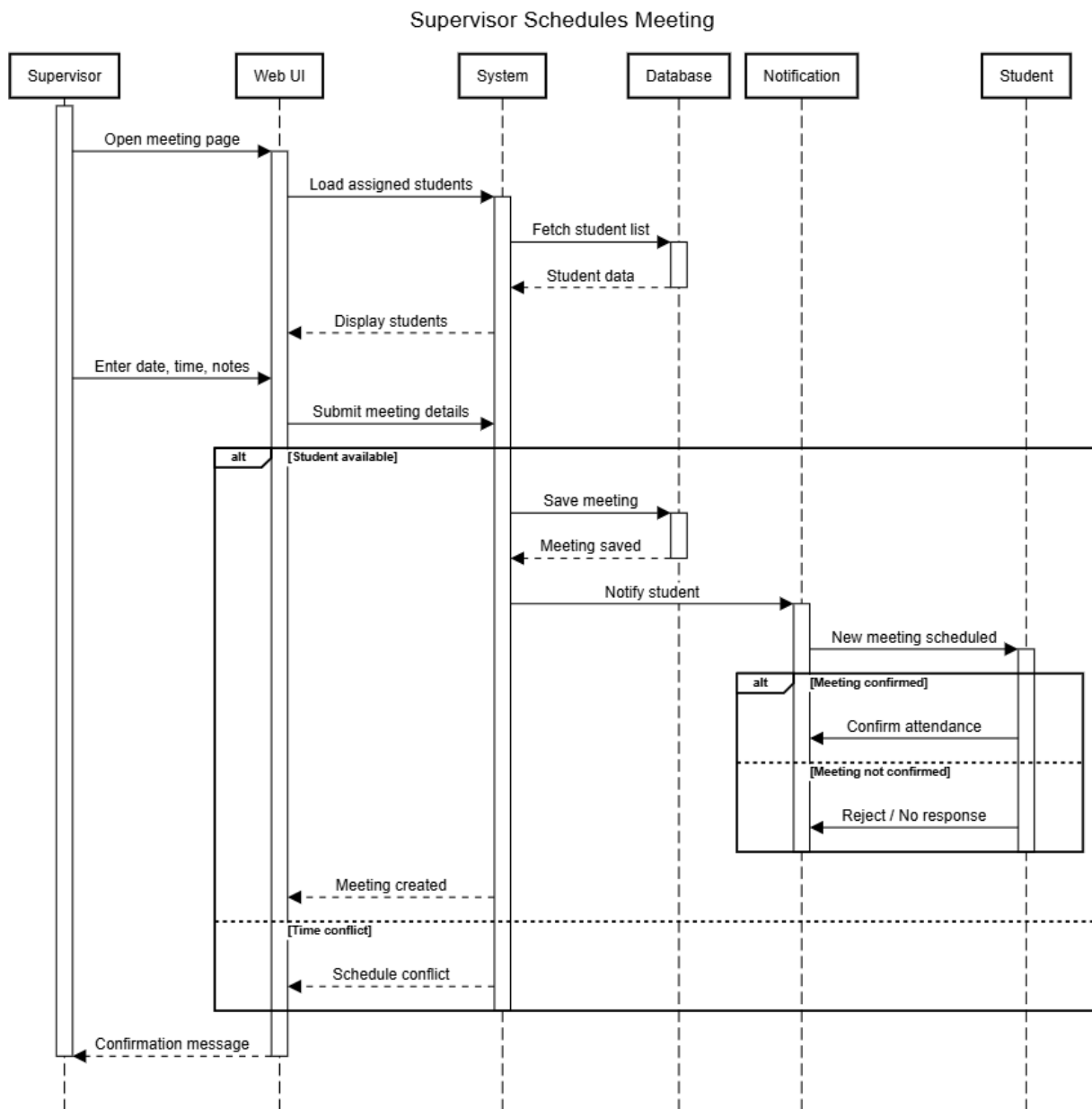


Figure 5: Sequence Diagram - Supervisor Schedules Meeting

3.Modeling

1. UML Use Case Diagram

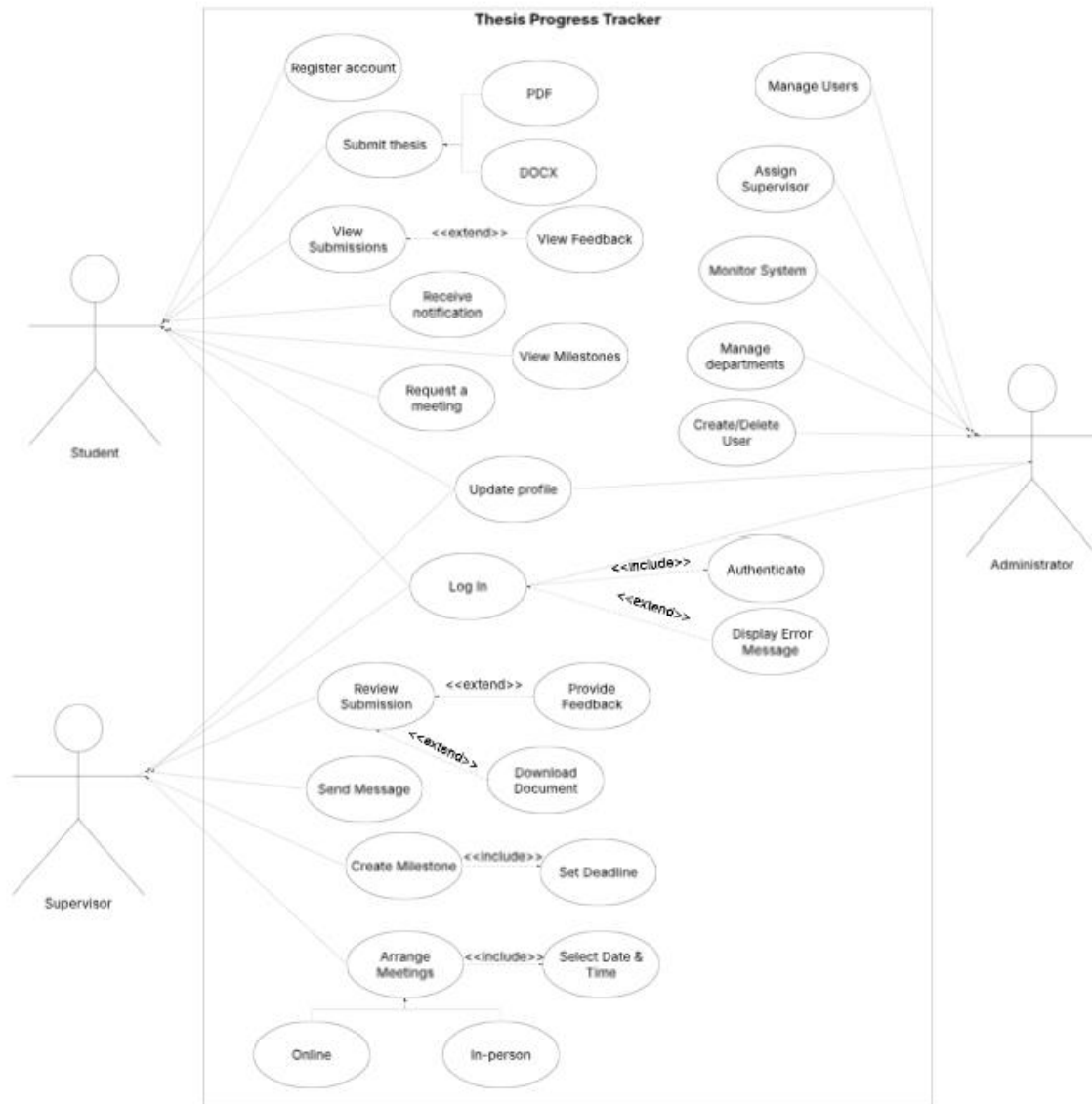


Figure 6: UML Use Case Diagram - Thesis Progress Tracker

(Note*: A clearer version of the diagram can be accessed via the following link)

Diagram Link: https://lucid.app/lucidchart/e4ac05c0-62ee-4867-a5b7-80cf090ee67d/edit?viewport_loc=1829%2C2364%2C5349%2C2456%2C0_0&invitationId=inv_90efec34-bc19-4739-8243-ca40f78ba04f

2. UML Activity Diagram

a. Thesis Submission Flow

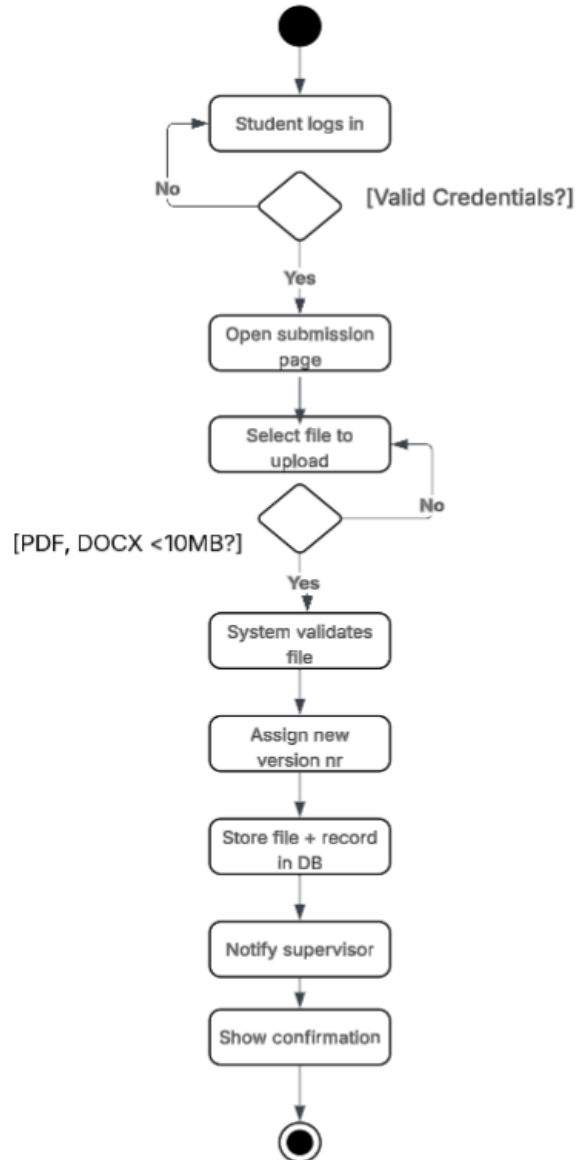


Figure 7: Activity Diagram - Thesis Submission

Diagram Link (for a & b): https://lucid.app/lucidchart/ab7938c5-5fb1-4d39-8021-d1e64c4f6cf3/edit?viewport_loc=1745%2C2176%2C2973%2C1365%2C0_0&invitationId=inv_42804748-2100-49cf-881f-6bda95a3e554

b. Supervisor Reviews Submission

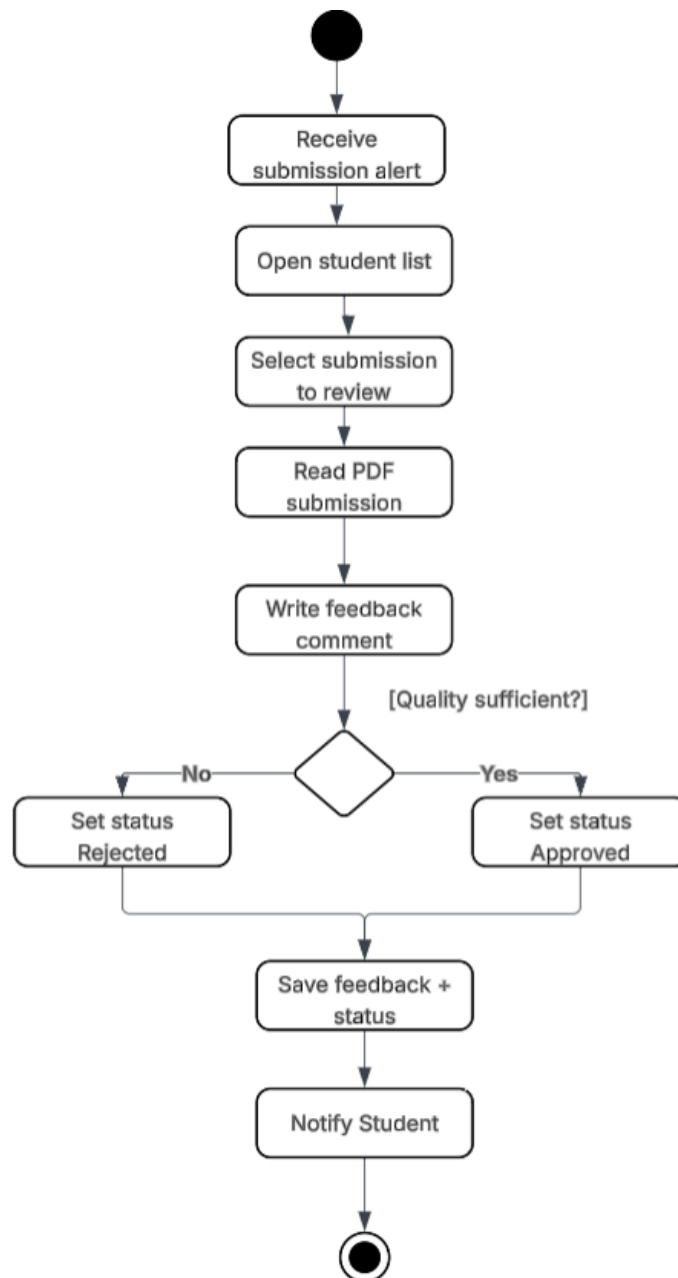


Figure 8: UML Activity Diagram - Reviewing Submission

3. UML State Diagram

a. Meeting Request

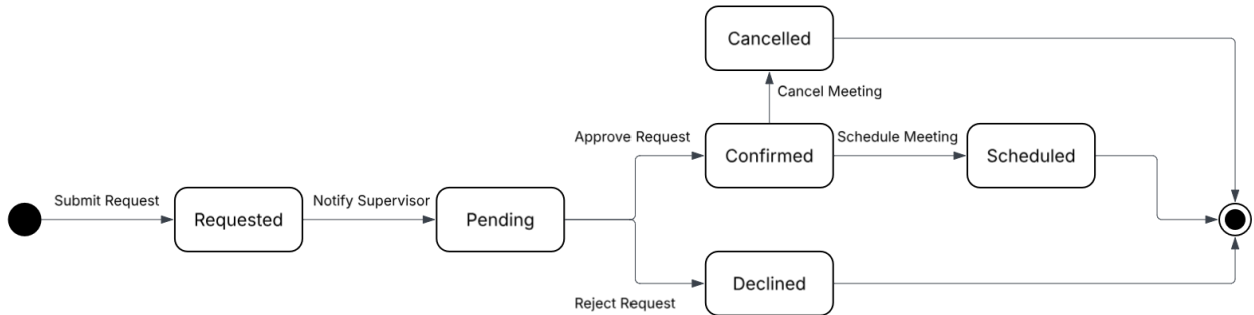


Figure 9: UML State Diagram – Meeting Request

b. File Submission

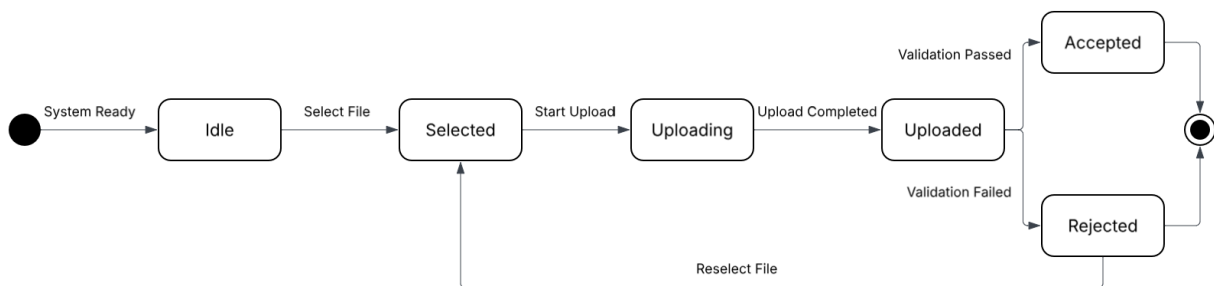


Figure 10: UML State Diagram - File Submission

(Note*: A clearer version of both state diagrams a & b can be accessed via the following link)

Diagram Link: https://lucid.app/lucidchart/ab7938c5-5fb1-4d39-8021-d1e64c4f6cf3/edit?viewport_loc=1745%2C889%2C2973%2C1365%2C0_0&invitationId=inv_42804748-2100-49cf-881f-6bda95a3e554